

Robotics & Automation · Multi-sensor System

Multi-sensor Registration for an Autonomous Forklift

Apply the coordinate registration used in surgical navigation between the robot, cameras, and sensors, and connect perception to safety decisions and the vehicle system.

Period 2025.11 - present | Evidence state Ongoing | Portfolio tier Robotics & Automation

Connected Jetson-based ToF-RGB vision (two-stage calibration, SAM3 segmentation, quality gates, continuous truck-load tracking), vision-servoing actions, a LiDAR sensor-integration service, and the initial safety-manager design into one integration flow on Zenoh-based in-house middleware.

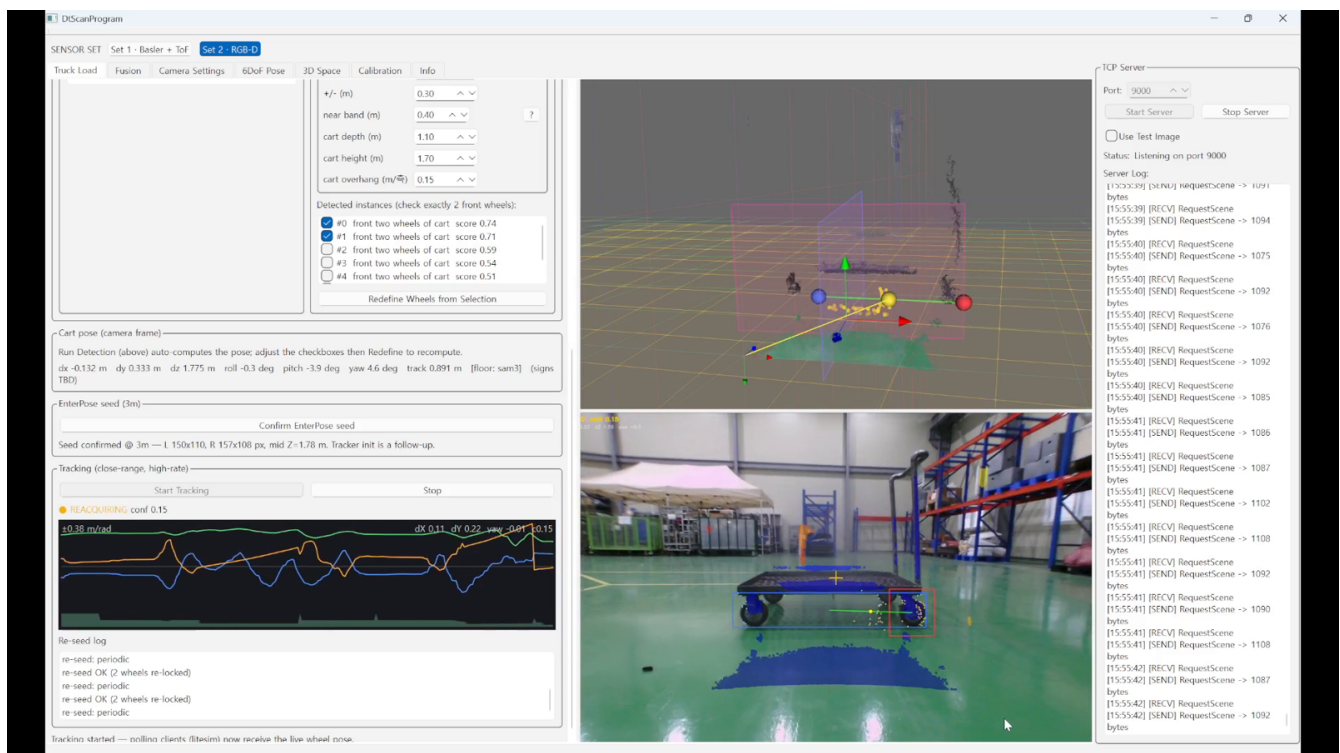


Figure 1. Screen recording of the DOTOR1 vision-servoing test (3D view, camera feed, signal plots).

Problem and system boundary

Sensor streams arrived in different coordinates and cycles and needed a common flow usable by safety decisions.

Owned decisions and implementation

My role

Designed and implemented the vision service on Jetson (two-stage ToF↔RGB calibration, a SAM3 segmentation sidecar, quality gates, continuous truck-load tracking) and the earlier Windows prototype. Implemented vision servoing, from wheel-pair detection to fork alignment, approach, and insertion, plus loading and unloading actions, and integrated them into the team's behaviour-tree framework. Built a sensor-integration service with a fused 3D occupancy output, mast-LiDAR surface detection and visualisation, and the service that publishes vision results over Zenoh; designed and initially implemented the safety manager (three-zone virtual bumper and E-STOP latching policy). Also contributed to the web-based operator and monitoring app UI and to Isaac Sim simulation work.

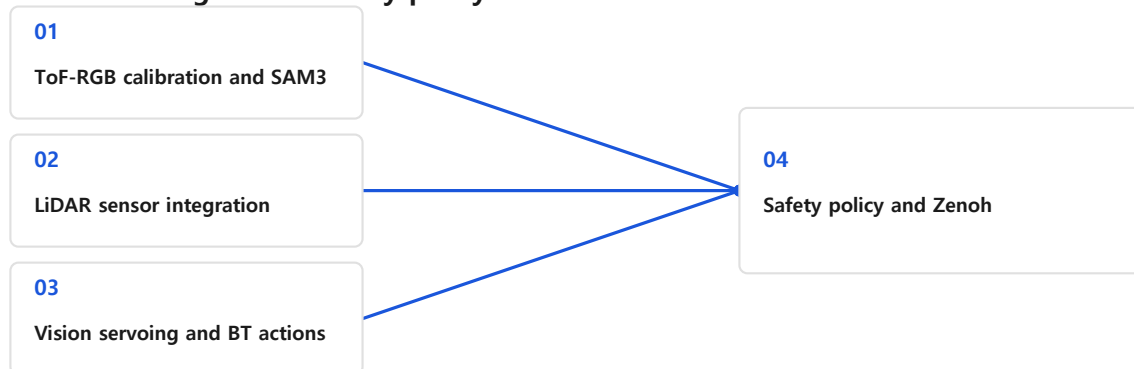
Sensor coordinate chain

Connected two-stage ToF↔RGB calibration and the mast LiDAR to vehicle coordinates, and produced one 3D occupancy output through the sensor-integration service.

Perception to policy

Published vision results over Zenoh, integrated vision-servoing and loading actions into the team's behaviour trees, and drew up the safety manager's initial three-zone virtual-bumper and E-STOP latching policy.

Multi-sensor convergence to safety policy



Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

Test-site recordings, photographs, and capture-specific sensor comparisons provide the evidence. The integration-evidence section below states the timestamp and extrinsic-measurement conditions and differences by axis.

Integration evidence

In a 191-frame, 9.7399-second capture on 29 July 2026, the ToF $\leftrightarrow$ RGB timestamp difference had a p95 of 16.592 ms (16.6 ms rounded). This is not end-to-end system latency. A single-frame extrinsic-translation comparison on 11 September was within 1 mm of the mechanical model in x and y, but differed by about 29 mm in z; it does not establish overall 3D accuracy.

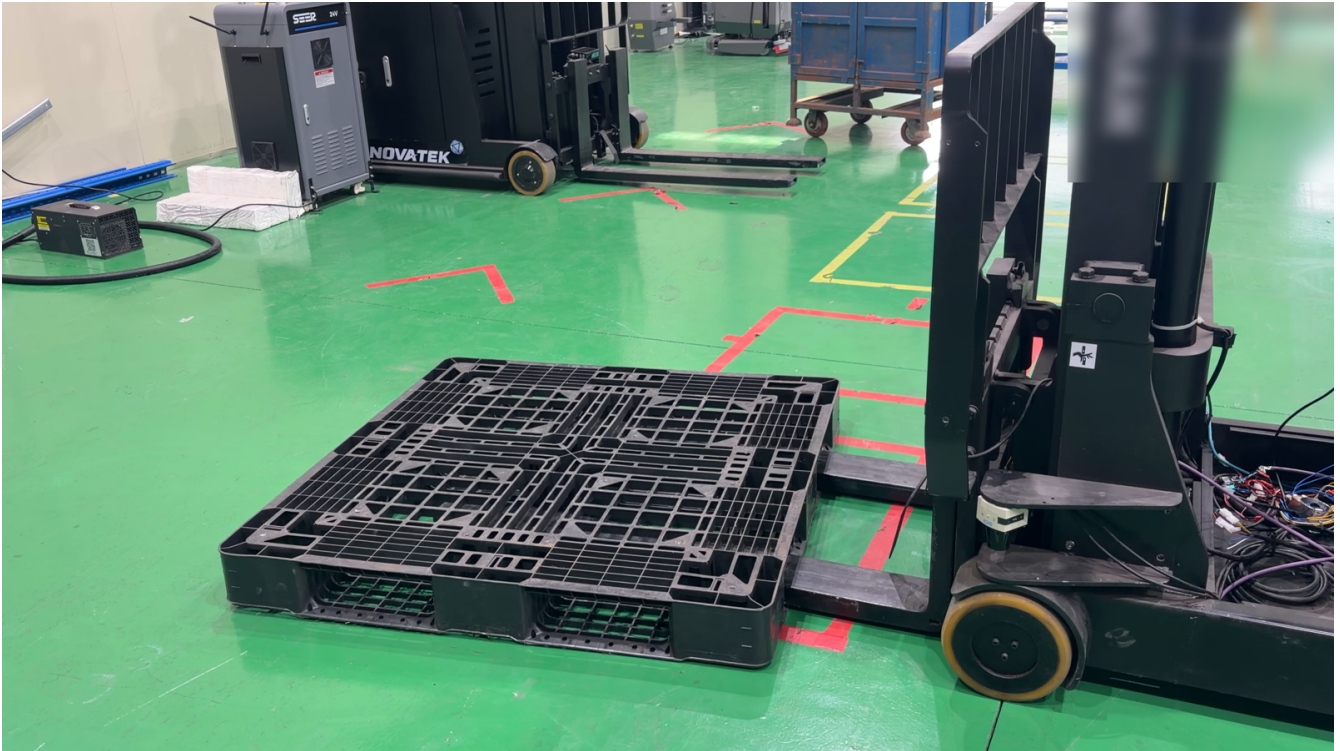


Figure 2. Fork alignment just before pallet entry during the full-scenario test



Figure 3. Trajectory-following drive test, wide view of the test site



Figure 4. Fork-lift test lifting a cart

Role, team result, and limits

Team result

Localisation (including NAV350), vehicle control and hardware abstraction, mission management, the behaviour-tree framework, the point-cloud publisher, and later safety-manager policy extensions were owned by team members; the team performed system integration and field validation. This is not presented as production operation, deployment success, or customer outcomes.

Limitations

The public clip and photos are test-site runs. The measurements are internal development measurements; no production or customer outcomes are claimed.

Field boundary

In earlier forklift field work during 2025, I supported operation, issue observation, and handovers. The public record of participation in the in-house DOTORI project begins in November 2025, followed by implementation records in the current repositories during 2026. Field support and later vision/sensor implementation are distinct; test-site validation is not a production or customer outcome.

Collaboration boundary

Vehicle-control, localisation, mission, safety, and field-validation owners integrate the system jointly. The middleware is an in-house Zenoh-based implementation that follows ROS 2-style patterns.

Implementation technologies

C++ · Zenoh · BehaviorTree.CPP · ToF · RGB · SAM3 · Jetson AGX Orin · SICK TiM LiDAR

Primary capability

Robot Vision & Sensor Integration · 3D Registration & Computer Vision

Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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